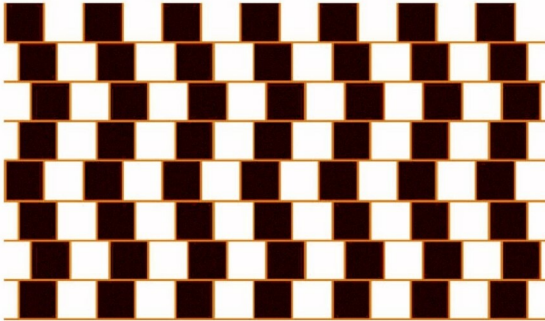


Get Smart With GNU Parallel

This article helps you understand the basics of GNU Parallel and its usage with examples



GNU Parallel is a tool for running jobs in parallel in a Bash environment. The job could be a single command or a script, with variable arguments. The simultaneous execution can occur on remote machines as well.

Released under the GPLv3+ licence, you can install it on Fedora using the following command:

```
$ sudo yum install parallel
```

After installation, you need to remove '-tollf' from the `/etc/parallel/config` file, if it is present. This option will be permanently removed in future releases.

GNU Parallel takes a command and a list of arguments for processing. The arguments are provided in the command line after the notation ' ::: ' and the command is executed for each argument. For example:

```
$ parallel echo ::: alpha beta gamma
alpha
beta
gamma
```

You can pass multiple arguments to GNU Parallel, and it will run the command for every combination of the input, as shown below:

```
$ parallel echo ::: 0 1 ::: 0 1
0 0
0 1
1 0
1 1
```

The order in the output may be different. The tool provides a number of replacement string options. The default string '{ }' represents the input:

```
$ parallel echo { } ::: /tmp
/tmp
```

The replacement string '{/}' removes everything up to and including the last forward slash:

```
$ parallel echo {/} ::: /tmp/stdio.h
stdio.h
```

If you want to return the path only, use the '{/}' string:

```
$ parallel echo {/} ::: /tmp/stdio.h
/tmp
```

The string '{.}' removes any filename extension: